OFFICE OF THE CYBER ARCHITECT SYSTEM MANUAL // SEC-REF: DORNAL-PW-WASM

SECRETUM CRYPTOGRAPHIC ENGINE HANDBOOK

High-Performance C++ WebAssembly Password Architecture Guide

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CORE: **C++ / WEBASSEMBLY (WASM)**

CLASSIFICATION: **100% SECURE CLIENT-SIDE**

EXECUTIVE ARCHITECTURAL SUMMARY

Secretum is an advanced, offline-first cryptographic password generator that ensures absolute key sovereignty. By compiling high-speed mathematical index mappings inside a **C++**

WebAssembly (WASM) binary sandbox and seeding it with hardware-derived entropy via standard JavaScript interfaces (`window.crypto.getRandomValues`), Secretum delivers uniform distribution results that are immune to classical interpretation profiling or local prediction vectors.

1. Cryptographic Entropy Pipeline

Standard software randomizers often generate predictable byte series over recurring loops. Secretum completely mitigates security degradation through a segmented execution model:

> **PHASE I: HARDWARE ENTROPY SEEDS**
> The browser accesses direct local CPU/kernel hardware entropy to generate high-degree 32-bit state integers through the native `window.crypto.getRandomValues` framework API.

> **PHASE II: HIGH-SPEED WEBASSEMBLY VIRTUAL MACHINE BOUNDARY**
> Raw entropy floats are securely transferred as binary inputs directly across the WASM runtime memory interface line. The compiled C++ binary executes division operations to map indexes deterministically without modulo bias.

COMPARATIVE ATTRIBUTE SPECIFICATIONS

Attribute	C++ WebAssembly Implementation Value
Virtualization Core	Client Sandbox Chrome/Safari WebAssembly Assembly instructions
Security Guarantee	No remote analytics transmission, 100% offline local memory cache

Attribute	C++ WebAssembly Implementation Value
Dictionary Modulo Bias	Mathematical zero-bias calculated natively inside fast pointers

2. Platform Portability & Standalone Setup Channels

📦 A. PACKAGED WEB APPLICATION EXTENSIONS (CHROMIUM SUITE)

Secretum can be loaded directly as a persistent browser extension:

1. Open your project and build: `npm run build` (prepares `manifest.json` in `/dist`).
2. Navigate command window directly to your Chrome Extensions suite: `chrome://extensions/`
3. Toggle active the **Developer Mode** checkbox in the top header.
4. Select **Load Unpacked** and import your compiled `/dist` directory directly.

📱 B. NATIVE PWA MOBILE PINS (APPLE IOS & GOOGLE ANDROID)

Featuring offline cache indexing (`sw.js`) and hardware app profiles (`site.webmanifest`), install Secretum directly on mobile screens:

APPLE IOS (SAFARI)

Open your deployed site link in Safari. Tap the share tray arrow icon, scroll, and select **Add to Home Screen**.

GOOGLE ANDROID (CHROME)

Open the site index in Chrome. Click install prompt banner, or tap the triple vertical menu dots and select **Install App**.